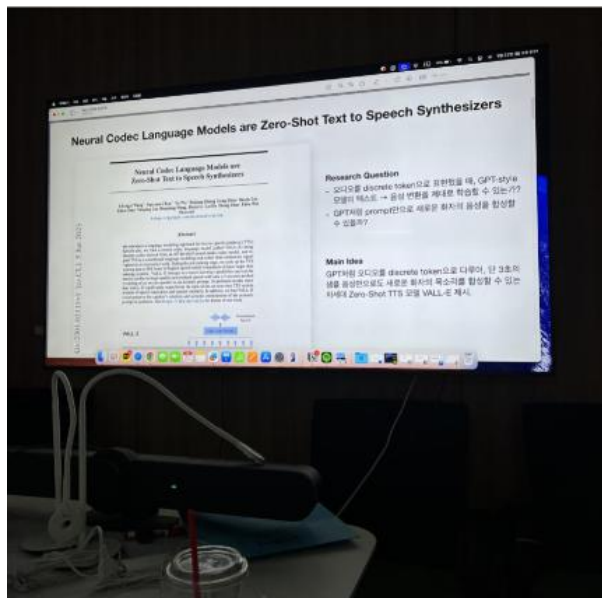


CAUI 딥러닝 논문 리뷰 스터디 (음성 처리)

2025.11.11

발표자 : 양희원

스터디원 소개 및 만남 인증



스터디원 1: 김동건

스터디원 2: 양희원

스터디원 3: 이나현

스터디 방식

월요일 15:00 – 16:30

대면 스터디 진행

각자 공부할 논문 선정 후 간단하게 PPT 제작 후
스터디에서 발표하며 공부한 논문의 내용을 공유

10월 13일 2차 스터디

10월 20일 3차 스터디

11월 2일 4차 스터디

스터디 내용

MATCHA-TTS: A FAST TTS ARCHITECTURE WITH CONDITIONAL FLOW MATCHING

Shivam Mehta, Ruibo Tu, Jonas Beskow, Éva Székely, Gustav Eje Henter

Division of Speech, Music and Hearing, KTH Royal Institute of Technology, Stockholm, Sweden

ABSTRACT

We introduce Matcha-TTS, a new encoder-decoder architecture for speedy TTS acoustic modelling, trained using optimal-transport conditional flow matching (OT-CFM). This yields an ODE-based decoder capable of high output quality in fewer synthesis steps than models trained using score matching. Careful design choices additionally ensure each synthesis step is fast to run. The method is probabilistic, non-autoregressive, and learns to speak from scratch without external alignments. Compared to strong pre-trained baseline models, the Matcha-TTS system has the smallest memory footprint, rivals the speed of the fastest model on long utterances, and attains the highest mean opinion score in a listening test.

Index Terms—Diffusion models, flow matching, speech synthesis, text-to-speech, acoustic modelling

1. INTRODUCTION

Diffusion probabilistic models (DPMs) (cf. [1]) are currently setting new standards in deep generative modelling on continuous and data-generation tasks such as image synthesis [2, 3], motion synthesis [4, 5], and speech synthesis [6, 7, 8, 9, 10] – the topic of this paper. DPMs define a diffusion process which transforms the *data* (a.k.a. *target*) distribution to a *prior* (a.k.a. *source*) distribution, e.g., a Gaussian. They then learn a *sampling process* that reverses the diffusion process. The two processes can be formulated as forward- and reverse-time SDE initial value problem generates samples from the learnt data distribution. Furthermore, each reverse-time SDE has a corresponding ordinary differential equation (ODE), called the *probability flow ODE* [11, 12], which describes (and samples from) the exact same distribution as the SDE. The probability flow ODE is a deterministic process for turning source samples into data samples, similar to continuous-time normalising flows (CNF) [13], but without the need to backpropagate through expensive ODE solvers or approximate the reverse ODE using adjoint variables [13].

The SDE formulation of DPMs is trained by approximating the score function (the gradients of the log probability density) of the data distribution [11]. The training objective takes the form of a mean squared error (MSE) which can be derived from an evidence lower bound (ELBO) on the likelihood. This is fast and simple and, unlike typical normalising flow models, does not impose any restric-

requires that an entire neural network be evaluated. This slow synthesis speed has long been the main practical issue with DPMs.

This paper introduces *Matcha-TTS*, a probabilistic and non-autoregressive, fast-to-sample-from TTS acoustic model based on continuous normalising flows. There are two main innovations:

1. To begin with, we propose an improved encoder-decoder TTS architecture that uses a combination of 1D CNNs and Transformers in the decoder. This reduces memory consumption and is fast to evaluate, improving synthesis speed.
2. Second, we train these models using optimal-transport conditional flow matching (OT-CFM) [14], which is a new method to learn ODEs that sample from a data distribution. Compared to conventional CNFs and score-matching probability flow ODEs, OT-CFM defines simpler paths from source to target, enabling accurate synthesis in fewer steps than DPMs.

Experimental results demonstrate that both innovations accelerate synthesis, reducing the trade-off between speed and synthesis quality. Despite being fast and lightweight, Matcha-TTS learns to speak and align without requiring an external aligner. Compared to strong pre-trained baseline models, Matcha-TTS achieves fast synthesis with better naturalness ratings. Audio examples and code are provided at <https://shivammehta25.github.io/Matcha-TTS/>.

2. BACKGROUND

2.1. Recent encoder-decoder TTS architectures

DPMs have been applied to numerous speech-synthesis tasks with impressive results, including waveform generation [6, 10] and end-to-end TTS [7]. Diff-TTS [9] was first to apply DPMs for acoustic modelling. Shortly after, Grad-TTS [8] conceptualised the diffusion process as an SDE. Although these models, and descendants like Fast Grad-TTS [15], are non-autoregressive, TorToSe [16] demonstrated DPMs in an autoregressive TTS model with quantised latents.

The above models – like many modern TTS acoustic models – use an encoder-decoder architecture with Transformer blocks in the encoder. Many models, e.g., FastSpeech 1 and 2 [17, 18], use sinusoidal position embeddings for positional dependencies. This has been found to generalise poorly to long sequences; cf. [19]. Glow-TTS [20], VITS [21], and Grad-TTS instead use relative positional embeddings [22]. Unfortunately, these treat inputs outside a short context window as a “bag of words”, often resulting in unnatural prosody. LinearSpeech [23] instead employed rotational position

TTS

Text-To-Speech

텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)

텍스트 → (정렬기 aligner) → 발음 타이밍 정보 → 음성 생성

연구 Motivation 느린 생성 속도, 발음 타이밍 오류, 외부 정렬기 의존

- 새로운 Non-autoregressive TTS 구조 제안
- OT-CFM (Optimal-Transport Conditional Flow Matching) 기반으로 기존 Diffusion보다 적은 단계로 고속 음성 생성
- 모델이 스스로 텍스트와 오디오(음향)의 대응 관계를 학습
- 빠르면서도 자연스러운 발음 품질 확보

스터디 내용

AudioLDM: Text-to-Audio Generation with Latent Diffusion Models

Haohe Liu^{*1} Zehua Chen^{*2} Yi Yuan¹ Xinhao Mei¹ Xubo Liu¹ Danilo Mandic²
Wenwu Wang¹ Mark D. Plumbley¹

Abstract

Text-to-audio (TTA) systems have recently gained attention for their ability to synthesize general audio based on text descriptions. However, previous studies in TTA have limited generation quality with high computational costs. In this study, we propose AudioLDM, a TTA system that is built on a latent space to learn continuous audio representations from contrastive language-audio pretraining (CLAP) embeddings. The pretrained CLAP models enable us to train LDMs with audio embeddings while providing text embeddings as the condition during sampling. By learning the latent representations of audio signals without modelling the cross-modal relationship, AudioLDM improves both generation quality and computational efficiency. Trained on AudioCaps with a single GPU, AudioLDM achieves state-of-the-art TTA performance compared to other open-sourced systems, measured by both objective and subjective metrics. AudioLDM is also the first TTA system that enables various text-guided audio manipulations (e.g., style transfer) in a zero-shot fashion. Our implementation and demos are available at <https://audioldm.github.io>.

1. Introduction

Generating sound effects, music, or speech according to personalized requirements is important for applications such as augmented and virtual reality, game development, and video editing. Traditionally, audio generation has been achieved through signal processing techniques (Andresen, 1979; Karplus & Strong, 1983). In recent years, gener-

ative models (Oord et al., 2016; Ho et al., 2020; Song et al., 2021; Tan et al., 2022), either unconditional or conditioned on other modalities (Kreuk et al., 2022; Żełaszczuk & Mańdziuk, 2022), have revolutionized this task. Previous studies primarily worked on the label-to-sound setting with a small set of labels (Liu et al., 2021b; Pascual et al., 2022) such as the ten sound classes in the UrbanSound8K dataset (Salamon et al., 2014). In comparison, natural language is considerably more flexible than labels as they can include fine-grained descriptions of audio signals, such as pitch, acoustic environment, and temporal order. The task of generating audio prompted with natural language descriptions is known as *text-to-audio* (TTA) generation.

TTA systems are designed to generate a wide range of high-dimensional audio signals. To efficiently model the data, we adopt a similar approach as DiffSound (Yang et al., 2022) by employing a learned discrete representation to efficiently model high-dimensional audio signals. We also draw inspiration from the recent advancements in autoregressive modelling of discrete representation learnt on the waveform, such as AudioGen (Kreuk et al., 2022), which has surpassed the capabilities of DiffSound. Building on the success of StableDiffusion (Rombach et al., 2022), which uses latent diffusion models (LDMs) for high-quality image generation, we extend previous TTA approaches to continuous latent representations, instead of learning discrete representations. Additionally, as audio manipulations, such as style transfer (Engel et al., 2020; Pascual et al., 2022), are desired for some applications such as games, we explore and achieve various zero-shot text-guided audio manipulations with LDMs, which have not been demonstrated before.

For previous TTA works, a potential limitation for generation quality is the requirement of large-scale high-quality audio-text data pairs, which are usually not readily available, and where they are available, are of limited quality and quantity (Liu et al., 2022f). To better utilize the low-quality data, several methods for text preprocessing have been pro-

TTS	Text-To-Speech	텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)
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ASR	Automatic Speech Recognition	STT와 거의 같은 뜻으로, 자동 음성 인식 전반을 가리킴
TTA	Text-To-Audio	TTS보다 범위가 넓은 개념으로, 말뿐 아니라 음악, 효과음 등 비언어적 오디오 생성까지 포함하는 경우도 있음

연구 Motivation

오디오 다양성 부족, 텍스트-음향 의미 연결 약함

- Text-to-Audio 생성 모델
- 텍스트를 CLAP 임베딩으로 오디오와 같은 잠재공간에 매핑
- VAE + Latent Diffusion Model 조합으로 다양한 오디오 생성
- 사운드 효과·음악·환경음 등 다양한 오디오 생성 가능
- 오디오 데이터만으로 학습 가능 (텍스트 없이도 전이 가능)

^{*}Equal contribution ¹Centre for Vision, Speech and Siz-

스터디 내용

AudioX: Diffusion Transformer for Anything-to-Audio Generation

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Xu Tan², Qifeng Chen¹, Wei Xue¹, Yike Guo¹

¹Hong Kong University of Science and Technology

²Independent Researcher

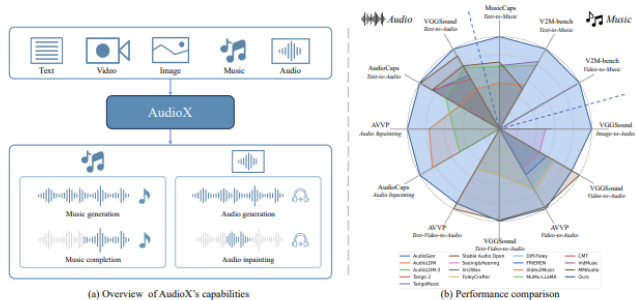


Figure 1. (a) Overview of AudioX, illustrating its capabilities across various tasks. (b) Radar chart comparing the performance of different methods across multiple benchmarks. AudioX demonstrates superior Inception Scores (IS) across a diverse set of datasets in audio and music generation tasks.

Abstract

Audio and music generation have emerged as crucial tasks in many applications, yet existing approaches face significant limitations: they operate in isolation without unified capabilities across modalities, suffer from scarce high-quality, multi-modal training data, and struggle to effectively integrate diverse inputs. In this work, we propose AudioX¹, a unified Diffusion Transformer model for Anything-to-Audio and Music Generation. Unlike previous domain-specific models, AudioX can generate both general audio and music with high quality, while offering flexible natural language control and seamless processing of various

modalities including text, video, image, music, and audio. Its key innovation is a multi-modal masked training strategy that masks inputs across modalities and forces the model to learn from masked inputs, yielding robust and unified cross-modal representations. To address data scarcity, we curate two comprehensive datasets: vggSound-caps with 190K audio captions based on the VGGSound dataset, and V2M dataset with 6 million music captions derived from the V2M dataset. Extensive experiments demonstrate that AudioX not only matches or outperforms state-of-the-art specialized models, but also offers remarkable versatility in handling diverse input modalities and generation tasks within a unified architecture. The code and datasets will be available at <https://zeyuet.github.io/AudioX/>.

¹Work in progress

TTS	Text-To-Speech	텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)
STT	Speech-To-Text	음성을 텍스트로 변환하는 기술 (예: 음성 인식 자막, Siri의 음성 입력)
ASR	Automatic Speech Recognition	STT와 거의 같은 뜻으로, 자동 음성 인식 전환을 가리킴
TTA	Text-To-Audio	TTS보다 범위가 넓은 개념으로, 말뿐 아니라 음악, 효과음 등 비언어적 오디오 생성까지 포함하는 경우도 있음

연구 Motivation

모달 간 표현 통합 미흡, 특정 모달 의존 심함

- 텍스트·이미지·비디오·오디오 → 오디오 생성 통합 모델
- Diffusion Transformer + 다중모달 마스킹으로 다른 입력 모달 간 embedding 학습 강화
- 텍스트→음악, 비디오→사운드 등 모든 변환 통합
- 입력 모달이 많을수록 성능 향상

스터디 내용

Fast Text-to-Audio Generation with Adversarial Post-Training

Zachary Novack^{1,2*}, Zach Evans^{2*}, Zack Zukowski², Josiah Taylor², CJ Carr², Julian Parker²
Adnan Al-Sinan², Gian Marco Iodice³, Julian McAuley³, Taylor Berg-Kirkpatrick³, Jordi Pons²

¹UC – San Diego, ²Stability AI, ³Arm, *Shared authorship, [†]Work done while an intern at Stability AI

Abstract—Text-to-audio systems, while increasingly performant, are slow at inference time, thus making their latency unpractical for many creative applications. We present Adversarial Relativistic-Contrastive (ARC) post-training, the first adversarial acceleration algorithm for diffusion/flow models not based on distillation. While past adversarial post-training methods have struggled to compare against their expensive distillation counterparts, ARC post-training is a simple procedure that (1) extends a recent relativistic adversarial formulation to diffusion/flow post-training and (2) combines it with a novel contrastive discriminator objective to encourage better prompt adherence. We pair ARC post-training with a number of optimizations to Stable Audio Open and build a model capable of generating ≈ 12 s of 44.1kHz stereo audio in ≈ 75 ms on an H100, and ≈ 7 s on a mobile edge-device, the fastest text-to-audio model to our knowledge.

1. INTRODUCTION

Despite the recent progress building generative text-to-audio models, such systems require seconds to minutes *per generation* [1]–[4], which limits their usability for most creative use cases. Our goal is to accelerate gaussian flow-based models, *i.e.*, diffusion models [5] or the more recent rectified flows [6]. Since gaussian flow models can be costly at inference time due to their iterative (step-based) sampling [7], considerable effort has gone into accelerating them [8]–[11]. Most of these works rely on (step) distillation, where the teacher model provides direct supervision to train a distilled few-step generator. As such, the few-step generator learns to map multiple inference steps into a single step, or a small number of steps, by distilling the teacher’s trajectories. However, most distillation approaches come with practical drawbacks: like *online* methods [9], [12]–[19], which are costly to train as they require 2–3 full models held in memory at the same time, or *offline* methods [20]–[22], which require significant resources to generate and store trajectory-output pairs offline to later train on. Furthermore, most distillation setups [12]–[15], [17], [18] distill the teacher with Classifier-Free Guidance (CFG), which is a double-edged sword, as the generator inherits both the high quality/prompt adherence as well as the low diversity and over-saturation artifacts of CFG [23].

To avoid such drawbacks, some studied acceleration through *post-training* (without distillation) [24], [25]. These works are primarily adversarial, as opposed to distillation methods that use adversarial auxiliary losses [9]–[11], [17], [18], and use real data rather than teacher-generated samples, thus freeing the costly requirement of using trajectory-output pairs. The motivation here is that the adversarial loss encourages *realism*, making each estimate better than the standard gaussian flow estimate. Such improved estimates enable post-trained models to use fewer (1–8) sampling steps [24], [25]. Recent image

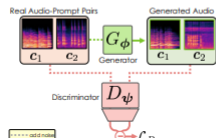


Fig. 1: ARC’s adversarial relativistic loss. Pairs of generated and real samples (with the same text prompts) are passed into the discriminator (with additive noise), where the generator and discriminator are trained to minimize and maximize (respectively) the difference between fake and real outputs.

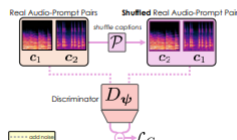


Fig. 2: ARC’s contrastive loss. The discriminator is also trained to maximize the difference between audios with correct and incorrect (shuffled) prompts.

on distillation [9]. Finally, most existing accelerated text-to-audio models focus on band-limited [26], mono [18], [27], [29] audio.

We show that rectified flows can be accelerated with adversarial post-training, introducing the first fully-adversarial text-to-audio acceleration framework that requires neither distillation nor CFG. We propose Adversarial Relativistic-Contrastive post-training (ARC) that (1) extends a recently proposed relativistic adversarial formulation [30], [31] to text-conditional audio generation and (2) combines this with a novel contrastive loss that encourages the discriminator to focus on prompt adherence. To our knowledge, we are the first to attempt such a post-training recipe on audio generation, the first to use adversarial relativistic losses for gaussian flow acceleration across modalities, and the first to use a contrastive discriminator formulation. We pair ARC with an array of architecture updates to Stable Audio Open (SAO) [4] to achieve accelerated performance, generating ≈ 12 s of 44.1kHz stereo audio in ≈ 75 ms on a H100 GPU, being 100x faster than the original

TTS	Text-To-Speech	텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)
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TTA	Text-To-Audio	TTS보다 범위가 넓은 개념으로, 말뿐 아니라 음악, 효과음 등 비언어적 오디오 생성까지 포함하는 경우도 있음

연구 Motivation

Diffusion/Flow 모델 추론 느림,
증류 시 품질 저하

- 고품질 TTA 모델의 추론 속도 문제 해결
- ARC(Adversarial Relativistic-Contrastive) 포스트트레이닝 제안
- 증류 없이 Diffusion/Flow 모델 속도 최대 100배 개선
- 품질 저하 없이 모바일 환경에서도 실시간 생성 가능

스터디 내용

Preprint

THE CURSE OF MULTI-MODALITIES: EVALUATING HALLUCINATIONS OF LARGE MULTIMODAL MODELS ACROSS LANGUAGE, VISUAL, AND AUDIO

Sicong Leng^{1,2*} Yun Xing^{2*} Zesen Cheng^{1,4*} Yang Zhou³ Hang Zhang^{1,4}
Xin Li^{1,4} Deli Zhao^{1,4} Shijian Lu^{2,†} Chunyan Miao² Lidong Bing^{1,4}

¹ DAMO Academy, Alibaba Group ² Nanyang Technological University
³ IHPC A*STAR, Singapore ⁴ Hupan Lab, Hangzhou, China

Project Page: cmm-damovl.site Github Repo: github.com/DAMO-NLP-SG/CMM

ABSTRACT

Recent advancements in large multimodal models (LMMs) have significantly enhanced performance across diverse tasks, with ongoing efforts to further integrate additional modalities such as video and audio. However, most existing LMMs remain vulnerable to hallucinations, the discrepancy between the factual multimodal input and the generated textual output, which has limited their applicability in various real-world scenarios. This paper presents the first systematic investigation of hallucinations in LMMs involving the three most common modalities: language, visual, and audio. Our study reveals two key contributors to hallucinations: overreliance on unimodal priors and spurious inter-modality correlations. To address these challenges, we introduce the benchmark *The Curse of Multi-Modalities (CMM)*, which comprehensively evaluates hallucinations in LMMs, providing a detailed analysis of their underlying issues. Our findings highlight key vulnerabilities, including imbalances in modality integration and biases from training data, underscoring the need for balanced cross-modal learning and enhanced hallucination mitigation strategies. Based on our observations and findings, we suggest potential research directions that could enhance the reliability of LMMs.

1 INTRODUCTION

Large Multi-modal Models (LMMs) have rapidly advanced, driving significant improvements across a wide range of tasks by effectively integrating and processing diverse data modalities. These models (Li et al., 2024a; Zhang et al., 2024; Hong et al., 2024; Yao et al., 2024; Wang et al., 2024a; Achiam et al., 2023; Team et al., 2023; Ormazabal et al., 2024), leveraging multimodal inputs such as image and text, have achieved notable performance gains, particularly in generating contextually accurate textual outputs. As the field evolves, there is a growing trend toward incorporating additional modalities, such as audio and video (Xu et al., 2024; Chen et al., 2024a; Wang et al., 2024c; Zhang et al., 2023; Cheng et al., 2024; Li et al., 2024b; Kong et al., 2024; Tang et al., 2023; Ghosh et al., 2024; Chu et al., 2024), to enhance LMMs' ability to understand and interact with complex real-world environments. However, despite these advancements, LMMs are prone to a critical issue known as hallucination, where the generated outputs do not accurately reflect the multimodal

TTS	Text-To-Speech	텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)
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연구 Motivation

모달 불균형으로 잘못된 정보 생성,
평가체계 부재

- 대형 멀티모달 모델(LMM) 환각(hallucination) 분석
- 텍스트·이미지·오디오 결합 시 생기는 오류 원인 규명
- 하나의 모달에 과도 의존 → 환각 발생
- 새 평가 데이터셋 **CMM** 제안
- 실제 객체 인지는 강하지만 “없는 것 거부” 능력은 약함

스터디 내용

Audio Mamba: Bidirectional State Space Model for Audio Representation Learning

Mehmet Hamza Erol*, Arda Senocak*, Jiu Feng, Joon Son Chung
Korea Advanced Institute of Science and Technology

* Equal Contribution

Abstract

Transformers have rapidly become the preferred choice for audio classification, surpassing methods based on CNNs. However, Audio Spectrogram Transformers (ASTs) exhibit quadratic scaling due to self-attention. The removal of this quadratic self-attention cost presents an appealing direction. Recently, state space models (SSMs), such as Mamba, have demonstrated potential in language and vision tasks in this regard. In this study, we explore whether reliance on self-attention is necessary for audio classification tasks. By introducing Audio Mamba (AuM), the first self-attention-free, purely SSM-based model for audio classification, we aim to address this question. We evaluate AuM on various audio datasets - comprising six different benchmarks - where it achieves comparable or better performance compared to well-established AST model. Code is available at <https://github.com/mhamzaerol/Audio-Mamba-AuM>

1. Introduction

In recent years, CNNs [15, 18] have been replaced with transformer-based architectures [7, 10, 26, 27] in a paradigm shift in deep learning, as transformers outperform convolutional neural networks. Not only does the performance of transformers exceed that of CNNs, but establishing a unified architecture among many different research fields and tasks — traditionally using completely different models — is another breakthrough [1, 2, 11, 12, 21, 23, 24, 29, 30]. Despite their success, transformers are hindered by their reliance on the computationally intensive self-attention mechanism. The $O(n^2)$ cost of attention is a natural concern when processing longer sequences. This limitation motivates the exploration of alternative architectures as a

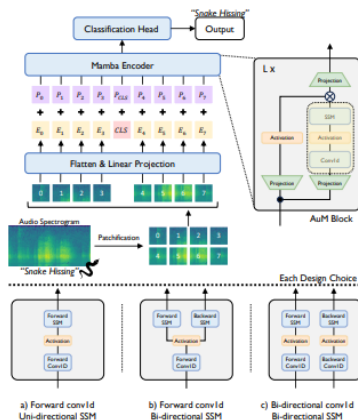


Figure 1. The proposed Audio Mamba (AuM) architecture.

tential alternative to transformer-based approaches. Given the universality and scalability of transformers across various tasks, Mamba's potential, coupled with its computational efficiency, is particularly promising for becoming a similarly generic and versatile architecture.

Despite Mamba's recent successes in language modeling and vision [19, 20, 31–33], the adoption of Mamba and similar SSM-based models in the audio classification

TTS	Text-To-Speech	텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)
STT	Speech-To-Text	음성을 텍스트로 변환하는 기술 (예: 음성 인식 자막, Siri의 음성 입력)
ASR	Automatic Speech Recognition	STT와 거의 같은 뜻으로, 자동 음성 인식 전반을 가리킴
TTA	Text-To-Audio	TTS보다 범위가 넓은 개념으로, 말뿐 아니라 음악, 효과음 등 비언어적 오디오 생성까지 포함하는 경우도 있음

연구 Motivation

Self-Attention 계산량 과다, 실시간 처리 어려움

- Self-Attention 없이 오디오 분류 수행
- 순수 상태공간모델(SSM) 기반 Audio Mamba (AuM) 제안
- AST의 quadratic 연산 복잡도 문제 해결
- Bidirectional SSM으로 문맥 이해력 강화
- 더 적은 연산으로 기존 Transformer급 성능

스터디 내용

Behavior-SD: Behaviorally Aware Spoken Dialogue Generation with Large Language Models

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<https://yhytoto12.github.io/Behavior-SD>

Abstract

Spoken dialogue involves behaviors like turn-taking, interruptions, filler words, and backchannels, which make interactions more natural and engaging but are often overlooked in language models. These models struggle to explicitly model these behavioral traits, resulting in a less natural and personalized communication style that aligns with user needs. To address this challenge, we make two key contributions. First, we introduce **Behavior-SD**, a large-scale dataset containing over 100K spoken dialogues (2,164 hours) annotated with various conversational behaviors, synthesized via LLMs to model diverse full-duplex interactions. Second, we propose **BeDLM**, the first dialogue model capable of generating natural conversations conditioned on specific behavioral and narrative contexts, supporting simultaneous contributions from both speakers. Through human evaluations and behavior-adherence metrics, we demonstrate that BeDLM outperforms baseline models in generating natural, coherent, and behaviorally rich dialogues. Our work opens new possibilities for developing behaviorally-aware dialogue systems that more closely mimic human conversational dynamics, enhancing user engagement and communication effectiveness.

1 Introduction

Large language models (LLMs) have significantly

els (Zhang et al., 2023; Li et al., 2024) struggle to replicate this complexity fully.

A particularly challenging aspect of real-world conversations is managing *full-duplex* communication, where speakers frequently overlap or interrupt each other. Full-duplex conversations, unlike traditional systems that enforce strict turn-taking, enable more fluid interactions, which can be beneficial in applications such as customer service, education, or voice assistant applications (Si et al., 2023; Google, 2024; OpenAI, 2024). Additionally, conversational dynamics are influenced by personal traits, social contexts, and factors like politeness (Bevacqua et al., 2012; Yamamoto et al., 2018), making this task even more complex.

Despite the prevalence of these dynamics in human conversations, they are often underexplored in dialogue generation. While existing spoken dialogue datasets (Cieri et al., 2004; Godfrey et al., 1992; Reece et al., 2023) include conversational behaviors such as interruptions and backchannels, these behaviors are not labeled at the speaker level, limiting the ability to model nuanced interaction patterns. Recent datasets (Lee et al., 2023; Lin et al., 2024) focus on sequential turn-taking, overlooking critical aspects of natural conversations like overlapping speech and interruptions.

To address these gaps, we present **Behavior-SD**, a large-scale dataset of over 100K spoken dialogues (SD), totaling 2,164 hours, and covering a wide range of conversational behaviors and social sit-

TTS	Text-To-Speech	텍스트를 음성으로 바꾸는 기술 (예: ChatGPT 음성 읽기, Google Assistant의 말하기 기능)
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연구 Motivation

비언어적 행동(추임새·끼어들기) 반영 안 됨

- 인간 대화의 행동적 요소(추임새, 끼어들기 등) 반영
- 대규모 음성 대화 데이터셋 **Behavior-SD(2,164시간)** 구축
- 화자 행동과 맥락을 조건으로 하는 **BeDLM** 모델 제안
- 자연스럽게 인간 같은 양방향 대화 생성 구현

스터디 내용

Robust Speech Recognition via Large-Scale Weak Supervision

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Abstract

We study the capabilities of speech processing systems trained simply to predict large amounts of transcripts of audio on the internet. When scaled to 680,000 hours of multilingual and multitask supervision, the resulting models generalize well to standard benchmarks and are often competitive with prior fully supervised results but in a zero-shot transfer setting without the need for any fine-tuning. When compared to humans, the models approach their accuracy and robustness. We are releasing models and inference code to serve as a foundation for further work on robust speech processing.

1. Introduction

Progress in speech recognition has been energized by the development of unsupervised pre-training techniques exemplified by WaveNet 2.0 (Baeck et al., 2020). Since these methods learn directly from raw audio without the need for human labels, they can productively use large datasets of unlabeled speech and have been quickly scaled up to 1,000,000 hours of training data (Zhang et al., 2021), far more than the 1,000 or so hours typical of an academic supervised dataset. When fine-tuned on standard benchmarks, this approach has improved the state of the art, especially in a low-data setting.

These pre-trained audio encoders learn high-quality representations of speech, but because they are purely unsupervised they lack an equivalently performant decoder mapping those representations to usable outputs, necessitating a fine-tuning stage in order to actually perform a task such as speech recognition¹. This unfortunately limits their usefulness and impact as fine-tuning can still be a complex process requiring a skilled practitioner. There is an additional risk with requiring fine-tuning. Machine learning

methods are exceedingly adept at finding patterns within a training dataset which boost performance on held-out data from the same dataset. However, some of these patterns are brittle and spurious and don't generalize to other datasets and distributions. In a particularly disturbing example, Radford et al. (2021) documented a 9.2% increase in object classification accuracy when fine-tuning a computer vision model on the ImageNet dataset (Russakovsky et al., 2015) without observing any improvement in average accuracy when classifying the same objects on seven other natural image datasets. A model that achieves "superhuman" performance when trained on a dataset can still make many basic errors when evaluated on another, possibly precisely because it is exploiting those dataset-specific quirks that humans are oblivious to (Geirhos et al., 2020).

This suggests that while unsupervised pre-training has improved the quality of audio encoders dramatically, the lack of an equivalently high-quality pre-trained decoder, combined with a recommended protocol of dataset-specific fine-tuning, is a crucial weakness which limits their usefulness and robustness. The goal of a speech recognition system should be to work reliably "out of the box" in a broad range of environments without requiring supervised fine-tuning of a decoder for every deployment distribution.

As demonstrated by Narayanan et al. (2018), Likhonaneenko et al. (2020), and Chan et al. (2021) speech recognition systems that are pre-trained in a supervised fashion across many datasets/domains exhibit higher robustness and generalize much more effectively to held-out datasets than models trained on a single source. These works achieve this by combining as many existing high-quality speech recognition datasets as possible. However, there is still only a moderate amount of this data easily available. SpeechSearch (Chan et al., 2021) mixes together 7 pre-existing datasets totalling 5,140 hours of supervision. While not insignificant, this is still tiny compared to the previously mentioned 1,000,000 hours of unlabeled speech data utilized in Zhang et al. (2021).

Google USM: Scaling Automatic Speech Recognition Beyond 100 Languages

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Abstract

We introduce the Universal Speech Model (USM), a single large model that performs automatic speech recognition (ASR) across 100+ languages. This is achieved by pre-training the encoder of the model on a large unlabeled multilingual dataset of 12 million (M) hours spanning over 300 languages, and fine-tuning on a smaller labeled dataset. We use multilingual pre-training with random-projection quantization and speech-text modality matching to achieve state-of-the-art performance on downstream multilingual ASR and speech-to-text translation tasks. We also demonstrate that despite using a labeled training set 1/7-th the size of that used for the Whisper model [1], our model exhibits comparable or better performance on both in-domain and out-of-domain speech recognition tasks across many languages.

1 Introduction

Recent advances in self-supervised learning have ushered in a new era for speech recognition. Whereas previous work focused mostly on improving the quality of monolingual models for mainstream languages, recent studies have increasingly turned to "universal" models [1-4]. These may take the form of a single model that performs well on multiple tasks [1, 2], or one that covers multiple domains [2, 3], or one that supports multiple languages [1, 5]. In this work, we explore the frontiers of language expansion. Our long-term goal is to train a universal ASR model that covers all the spoken languages in the world.

- **Whisper:** 68만 시간 다언어 음성-텍스트 데이터 학습
→ 잡음 환경에서도 **강인한 음성 인식 모델**
- **USM:** 12M h 다국어 음성으로 학습한 **범용 ASR 모델**
→ 100개 이상 언어 처리, 저자원 언어에서도 강력
- Whisper 대비 **1/7 데이터로 동급 성능, 범용성 높음**